PROLOGUE

Don’t Be Afraid

You don’t need a PhD in theoretical physics to be afraid of quantum mechanics. But it doesn’t hurt.

That might seem strange. Quantum mechanics is our best theory of the microscopic world. It describes how atoms and particles interact through the forces of nature, and makes incredibly precise experimental predictions. To be sure, quantum mechanics has something of a reputation for being difficult, mysterious, just this side of magic. But professional physicists, of all people, should be relatively comfortable with a theory like that. They are constantly doing intricate calculations involving quantum phenomena, and building giant machines dedicated to testing the resulting predictions. Surely we’re not suggesting that physicists have been faking it all this time?

They haven’t been faking, but they haven’t exactly been honest with themselves either. On the one hand, quantum mechanics is the heart and soul of modern physics. Astrophysicists, particle physicists, atomic physicists, laser physicists—everyone uses quantum mechanics all the time, and they’re very good at it. It’s not just a matter of esoteric research. Quantum mechanics is ubiquitous in modern technology. Semiconductors, transistors, microchips, lasers, and computer memory all rely on quantum mechanics to function. For that matter, quantum mechanics is necessary to make sense of the most basic features of the world around us. Essentially all